

DEVILS LAKE, ND, USA

WMO: 727573

Lat: 48.117N

Lon: 98.900W

Elev: 1441

StdP: 13.95

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-16.8	-27.4	1.3	-16.0	-25.8	1.4	-14.1	33.2	12.4	30.8	11.6	8.0	260	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	86.3	70.0	82.5	67.5	80.9	66.3	72.8	82.2	70.4	80.3	68.3	77.8	12.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.9	116.1	79.5	66.3	102.2	76.8	64.2	94.9	74.3	37.4	82.3	35.2	80.4	33.5	78.1	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.9	25.9	23.3	-25.9	92.6	5.8	3.2	-30.2	94.9	-33.6	96.8	-36.9	98.6	-41.1	100.9	
			-24.6	76.6	4.3	2.8	-27.7	78.6	-30.2	80.2	-32.6	81.8	-35.7	83.8	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	7.5	10.6	24.1	39.6	53.2	63.6	68.8	67.0	58.1	43.2	27.6	13.0
	DBStd	24.17	14.57	13.64	13.22	10.67	8.69	6.69	5.65	6.08	8.32	9.24	11.26	13.23
	HDD50	5725	1319	1104	805	336	67	2	0	0	22	249	673	1148
	HDD65	9539	1784	1524	1269	763	376	102	28	48	235	675	1122	1613
	CDD50	2020	0	0	2	23	167	410	583	528	266	40	1	0
	CDD65	356	0	0	0	0	10	60	146	111	29	0	0	0
	CDH74	2837	0	0	0	5	122	464	1102	875	256	13	0	0
	CDH80	697	0	0	0	0	24	96	276	242	58	1	0	0
	WSAvg	11.3	11.4	11.3	11.5	11.9	12.2	10.9	9.8	9.9	10.8	12.2	11.9	11.4
	PrecAvg	17.4	0.6	0.5	0.9	1.1	2.2	3.3	2.6	2.1	1.9	1.1	0.7	0.6
Precipitation	PrecMax	27.8	1.3	1.4	3.2	3.5	6.3	6.6	8.1	4.2	5.2	4.9	3.9	1.2
	PrecMin	10.0	0.0	0.0	0.2	0.0	0.3	0.9	0.3	0.2	0.2	0.0	0.0	0.1
	PrecStd	3.8	0.4	0.4	0.7	0.7	1.5	1.6	1.7	1.1	1.3	1.1	0.7	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	41.4	61.0	72.6	83.0	87.7	90.4	90.8	86.0	75.0	60.2	41.4
		MCWB	34.4	36.4	49.5	53.5	62.2	68.9	72.8	72.5	68.5	58.9	48.9	35.5
	2%	DB	36.0	36.9	52.0	66.1	77.2	82.1	86.0	85.9	81.0	66.5	52.2	37.0
		MCWB	33.1	33.9	44.9	50.6	58.7	66.2	71.5	69.4	64.7	54.6	44.0	33.1
	5%	DB	32.4	34.3	45.5	61.3	72.6	79.4	82.4	82.0	76.4	62.6	46.3	34.2
		MCWB	30.5	31.9	39.9	47.7	56.2	64.0	68.7	67.4	62.0	51.3	40.1	31.4
	10%	DB	27.9	30.2	39.3	55.4	68.6	75.4	80.7	79.2	72.3	56.9	42.8	30.1
		MCWB	26.2	28.7	35.8	45.2	54.3	62.5	67.5	65.6	60.4	48.6	37.3	28.4
	0.4%	WB	34.9	37.4	51.3	55.1	66.2	73.3	76.4	75.5	70.5	61.0	49.8	36.6
		MCDB	37.8	40.2	56.9	69.4	77.6	83.2	85.7	86.1	81.1	71.2	58.9	41.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.0	34.1	44.9	51.5	62.1	69.3	73.4	71.7	67.3	56.1	44.5	33.6
		MCDB	35.6	37.0	51.5	63.8	72.4	79.3	82.1	81.8	77.0	65.5	51.0	36.8
	5%	WB	30.4	32.0	40.0	49.3	58.7	66.9	71.1	69.3	64.3	52.4	40.8	31.4
		MCDB	32.2	34.3	45.1	60.0	68.3	75.9	80.4	79.9	73.5	60.9	46.3	34.1
	10%	WB	26.4	28.5	36.2	46.2	56.1	64.3	69.0	67.4	61.7	49.3	37.3	28.3
		MCDB	28.0	30.2	40.1	55.1	66.1	73.1	78.4	76.7	70.2	56.9	42.2	29.9
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.6	15.9	18.5	19.7	17.5	18.0	19.1	19.1	16.2	14.3	14.3
		MCDBR	18.1	16.6	21.6	26.9	25.7	22.0	20.9	22.6	25.1	25.1	20.8	17.2
		MCWBR	16.3	14.2	15.2	14.6	12.7	10.7	10.6	11.2	12.9	14.8	14.1	14.3
	5% WB	MCDBR	17.9	15.8	21.4	24.9	21.4	19.0	19.1	20.7	21.9	23.2	20.4	16.9
		MCWBR	16.3	13.6	15.5	14.5	11.9	11.2	11.2	11.6	13.0	14.5	14.3	14.4
Clear-Sky Solar Irradiance	taub		0.286	0.292	0.294	0.361	0.373	0.377	0.381	0.384	0.351	0.297	0.268	0.251
	taud		2.252	2.278	2.443	2.310	2.341	2.378	2.378	2.355	2.425	2.552	2.498	2.430
	Ebn at Noon		246	274	296	282	282	281	278	272	271	274	260	254
	Edh at Noon		24	30	30	38	38	37	36	36	30	23	19	18
All-Sky Solar Radiation	RadAvg		512	875	1250	1569	1789	1942	2049	1702	1259	787	470	358
	RadStd		41	85	110	124	115	134	105	115	83	98	65	39

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page